

AA 203: Optimal and Learning-based Control

Homework #1

Due Friday, April 24 by 5:00 pm

Learning goals for this problem set:

Problem 1: To gain insights into the implementation of gradient methods and review some notions of linear algebra.

Problem 2: To familiarize with Linear Quadratic control and learn a first algorithmic approach to this problem.

Problem 3: Become familiar with the process of solving calculus of variations problems.

Problem 4: To familiarize with the Hamiltonian equations for optimal control.

1.1 Gradient descent and line search. Let $Q \in \mathbb{R}^{n \times n}$ be a symmetric positive-definite matrix, and $b \in \mathbb{R}^n$ be a given vector. Consider the quadratic optimization problem

$$\min_{x \in \mathbb{R}^n} \frac{1}{2} x^\top Q x - b^\top x.$$

Let $f(x) := \frac{1}{2} x^\top Q x - b^\top x$, and denote the eigenvalues of Q as $\lambda_1, \dots, \lambda_n$.

(a) Find the unique local minimum candidate $x^* \in \mathbb{R}^n$. Prove x^* is a global minimum.

Hint: Any twice-differentiable function f is strictly convex if the Hessian $\nabla^2 f(x)$ is positive-definite for all $x \in \mathbb{R}^n$.

(b) Show that, starting from any initial point $x^{(0)} \in \mathbb{R}^n$, Newton's method with constant step size $\eta = 1$ converges in one iteration to the optimal solution x^* . Hence, performing one step of Newton's method is equivalent to solving the linear system of equations $Qx = b$. What would be the downside of this solution method if n is large (e.g., $n \gg 10^4$) and the matrix Q has no particular structure?

(c) Let $S \in \mathbb{R}^{n \times n}$ be a symmetric matrix. By the Spectral Theorem, there exist an orthogonal matrix $U \in \mathbb{R}^{n \times n}$ and a diagonal matrix $\Sigma = \text{diag}(\mu_1, \dots, \mu_n)$ such that $S = U \Sigma U^\top$. Show $\|Sx\|_2 = \|\Sigma U^\top x\|_2$ for any $x \in \mathbb{R}^n$. Then show $\|\Sigma z\|_2 \leq \max_{i \in \{1, \dots, n\}} |\mu_i| \|z\|_2$ for any $z \in \mathbb{R}^n$. Finally, conclude that $\|Sx\|_2 \leq \max_{i \in \{1, \dots, n\}} |\mu_i| \|x\|_2$ for any $x \in \mathbb{R}^n$.

Hint: If $U \in \mathbb{R}^{n \times n}$ is an orthogonal matrix, then $\|Uy\|_2 = \|U^\top y\|_2 = \|y\|_2$ for any $y \in \mathbb{R}^n$.

(d) For any $\eta > 0$, show that the eigenvalues of the matrix $I - \eta Q$ are exactly $\{1 - \eta \lambda_i\}_{i=1}^n$.

Hint: Identify an orthonormal basis of vectors $\{v_i\}_{i=1}^n \subset \mathbb{R}^n$ such that $(I - \eta Q)v_i = (1 - \eta \lambda_i)v_i$ for each i .

(e) Consider the gradient descent update rule $x^{(k+1)} = x^{(k)} - \eta \nabla f(x^{(k)})$ at iteration $k \in \mathbb{N}_{\geq 0}$ with a constant step size $\eta > 0$. Define $\delta^{(k)} := \|x^{(k)} - x^*\|_2$ and $\gamma(\eta) := \max_{i \in \{1, \dots, n\}} |1 - \eta \lambda_i|$. Use an inductive argument to show $\delta^{(k)} \leq \gamma(\eta)^k \delta_0$ for all $k \in \mathbb{N}_{\geq 0}$.

(a) Find the unique local minimum candidate $x^* \in \mathbb{R}^n$. Prove x^* is a global minimum.

Hint: Any twice-differentiable function f is strictly convex if the Hessian $\nabla^2 f(x)$ is positive-definite for all $x \in \mathbb{R}^n$.

$$f(x) = \frac{1}{2} x^T Q x - b^T x \quad \nabla f(x) \in \mathbb{R}^n$$

$$\nabla f(x) = Qx - b$$

$$\nabla^2 f(x) = Q \quad f(x) \text{ is strictly convex as } Q \text{ is PD}$$

Then the local minimum is the global minimum

$$\nabla f(x) = Qx - b = 0 \quad \therefore \quad x^* = Q^{-1}b$$

(b) Show that, starting from any initial point $x^{(0)} \in \mathbb{R}^n$, Newton's method with constant step size $\eta = 1$ converges in one iteration to the optimal solution x^* . Hence, performing one step of Newton's method is equivalent to solving the linear system of equations $Qx = b$. What would be the downside of this solution method if n is large (e.g., $n \gg 10^4$) and the matrix Q has no particular structure?

$$x^{(k+1)} = x^{(k)} - \eta (\nabla^2 f(x^{(k)}))^{-1} \nabla f(x^{(k)})$$

$$\text{with } \eta = 1$$

$$x^{(k+1)} = x^{(k)} - Q^{-1} [Qx^{(k)} - b] = x^{(k)} - x^{(k)} + Q^{-1}b = Q^{-1}b$$

$$x^{(k+1)} = x^*$$

$$\text{Starting from } x^{(0)} \rightarrow x^{(1)} = x^*$$

If $n \gg 10^4$ and Q has no structure, then solving the linear system can be expensive

(c) Let $S \in \mathbb{R}^{n \times n}$ be a symmetric matrix. By the Spectral Theorem, there exist an orthogonal matrix $U \in \mathbb{R}^{n \times n}$ and a diagonal matrix $\Sigma = \text{diag}(\mu_1, \dots, \mu_n)$ such that $S = U\Sigma U^T$. Show $\|Sx\|_2 = \|\Sigma U^T x\|_2$ for any $x \in \mathbb{R}^n$. Then show $\|\Sigma z\|_2 \leq \max_{i \in \{1, \dots, n\}} |\mu_i| \|z\|_2$ for any $z \in \mathbb{R}^n$. Finally, conclude that $\|Sx\|_2 \leq \max_{i \in \{1, \dots, n\}} |\mu_i| \|x\|_2$ for any $x \in \mathbb{R}^n$.

Hint: If $U \in \mathbb{R}^{n \times n}$ is an orthogonal matrix, then $\|Uy\|_2 = \|U^T y\|_2 = \|y\|_2$ for any $y \in \mathbb{R}^n$.

$$\textcircled{1} \quad \|Sx\|_2 = \|U\Sigma U^T x\|_2 = \|\Sigma U^T x\|_2 \quad (\text{as } U \text{ is orthogonal})$$

$$\textcircled{2} \quad \|\Sigma z\|_2 = \|\Sigma\|_2 \|z\|_2 = \sqrt{\sum_{i,j} \Sigma_{ij}^2} \sqrt{\sum_i z_i^2} = \sqrt{\sum_i \mu_i^2} \cdot \sqrt{\sum_i z_i^2} = \sqrt{\sum_i \mu_i^2 z_i^2}$$

Considering $\mu_i^2 \leq \max_i |\mu_i|^2 \quad \forall i \in \{1, \dots, n\}$, then

$$\|\Sigma z\|_2 = \sqrt{\sum_i \mu_i^2 z_i^2} \leq \sqrt{\sum_i (\max_i |\mu_i|)^2 z_i^2} = \max_i |\mu_i| \sqrt{\sum_i z_i^2} = \max_i |\mu_i| \|z\|_2$$

$$\textcircled{3} \quad \|Sx\|_2 = \|\Sigma U^T x\|_2 = \|\Sigma x\|_2.$$

$$\text{Then} \quad \|\Sigma x\|_2 \leq \max_i |\mu_i| \|x\|_2$$

$$\text{Finally} \quad \|Sx\|_2 \leq \max_i |\mu_i| \|x\|_2$$

(d) For any $\eta > 0$, show that the eigenvalues of the matrix $I - \eta Q$ are exactly $\{1 - \eta \lambda_i\}_{i=1}^n$.

Hint: Identify an orthonormal basis of vectors $\{v_i\}_{i=1}^n \subset \mathbb{R}^n$ such that $(I - \eta Q)v_i = (1 - \eta \lambda_i)v_i$ for each i .

Since Q is symmetric, it has an orthonormal basis of eigenvectors v_i , $i \in \{1, \dots, n\}$

Then for each eigenvector v_i , $Qv_i = \lambda_i v_i$

$$\text{So} \quad (I - \eta Q)v_i = v_i - \eta Qv_i = v_i - \eta \lambda_i v_i = (1 - \eta \lambda_i)v_i \quad \forall i \in \{1, \dots, n\}$$

(e) Consider the gradient descent update rule $x^{(k+1)} = x^{(k)} - \eta \nabla f(x^{(k)})$ at iteration $k \in \mathbb{N}_{\geq 0}$ with a constant step size $\eta > 0$. Define $\delta^{(k)} := \|x^{(k)} - x^*\|_2$ and $\gamma(\eta) := \max_{i \in \{1, \dots, n\}} |1 - \eta \lambda_i|$. Use an inductive argument to show $\delta^{(k)} \leq \gamma(\eta)^k \delta_0$ for all $k \in \mathbb{N}_{\geq 0}$.

Consider $x^* = Q^{-1}b$, so $b = Qx^*$

Then the gradient is

$$\nabla f(x^{(k)}) = Qx^{(k)} - b = Qx^{(k)} - Qx^* = Q(x^{(k)} - x^*)$$

$$\text{Therefore} \quad x^{(k+1)} = x^{(k)} - \eta Q(x^{(k)} - x^*)$$

$$\text{Then} \quad x^{(k+1)} - x^* = x^{(k)} - x^* - \eta Q(x^{(k)} - x^*) = (I - \eta Q)(x^{(k)} - x^*)$$

From $\textcircled{d}$, eigenvalues of $(I - \eta Q)$ are $(1 - \eta \lambda_i)$ $\forall i \in \{1, \dots, n\}$, so

$$\text{RHS} \Rightarrow \|I - \eta Q\|_2 \|x^{(k)} - x^*\| = \|I - \eta Q\|_2 \delta^{(k)} = \max_i |1 - \eta \lambda_i| \delta^{(k)}$$

Then, leveraging $\textcircled{c}$ $S = U \Sigma U^T$

where $S = I - \eta Q$, a symmetric matrix,

$U = [v_1 \dots v_n]$ is an orthogonal matrix of eigenvectors,

$\Sigma = \text{diag}(1 - \eta \lambda_1, \dots, 1 - \eta \lambda_n)$ is a diag. matrix of real eigenvalues

$$\text{then} \quad \|S(x^{(k)} - x^*)\|_2 = \|I - \eta Q\|_2 \delta^{(k)}, \text{ so}$$

$$\|I - \eta Q\|_2 \delta^{(k)} \leq \max_i |1 - \eta \lambda_i| \delta^{(k)} = \gamma(\eta) \delta^{(k)}$$

Finally $\|x^{(k+1)} - x^*\|_2 = \delta^{(k+1)} = \|I - \eta Q\|_2 \delta^{(k)} \leq \gamma(\eta) \delta^{(k)}$

$$\underline{\delta^{(k+1)} \leq \gamma(\eta) \delta^{(k)}}$$

ITERs

For $k=0$

$$\delta^{(1)} \leq \gamma(\eta) \delta^{(0)}$$

For $k=1$

$$\delta^{(2)} \leq \gamma(\eta) \delta^{(1)} \leq \gamma(\eta) (\gamma(\eta) \delta^{(0)}) = \gamma(\eta)^2 \delta^{(0)}$$

For $k=3$

$$\delta^{(3)} \leq \gamma(\eta) [\gamma(\eta) (\gamma(\eta) \delta^{(0)})] = \gamma(\eta)^3 \delta^{(0)}$$

$\vdots$

$k=N$

$$\delta^{(N+1)} \leq \gamma(\eta)^{N+1} \delta^{(0)}$$

So

$$\delta^{(k+1)} \leq \gamma(\eta)^{k+1} \delta^{(0)} \quad \forall k \in \mathbb{N}_{\geq 0}$$

- (f) Consider gradient descent with exact line search. At each iteration k , denote the descent direction by $d^{(k)} := -\nabla f(x^{(k)})$ and the optimal step size by

$$\eta^{(k)} := \arg \min_{\eta \geq 0} f(x^{(k)} + \eta d^{(k)}).$$

Prove

$$\eta^{(k)} = \frac{\|d^{(k)}\|_2^2}{d^{(k)\top} Q d^{(k)}}.$$

- (g) For $n = 2$ and $f(x) = \frac{1}{2}(x_1^2 + \gamma x_2^2)$ with $\gamma = 10$, what is the optimal solution x^* ? Implement gradient descent with a constant step size and exact line search, starting from $x^{(0)} = (5, 1)$ and $x^{(0)} = (1, 5)$. What do you observe with exact line search? When does gradient descent begin to “zig-zag”? What issue do you observe with a constant step size? Repeat both experiments with $\gamma = 1$. Submit your plots.

$$\begin{aligned} f) \quad f(x^{(k)} + \eta d^{(k)}) &= \frac{1}{2} (x^{(k)} + \eta d^{(k)})^\top Q (x^{(k)} + \eta d^{(k)}) - b^\top (x^{(k)} + \eta d^{(k)}) \\ &= \frac{1}{2} \left[x^{(k)\top} Q x^{(k)} + \eta x^{(k)\top} Q d^{(k)} + \eta d^{(k)\top} Q x^{(k)} + \eta^2 d^{(k)\top} Q d^{(k)} \right] - b^\top x^{(k)} - \eta b^\top d^{(k)} \end{aligned}$$

Q is symmetric, so

$$\begin{aligned} f(x^{(k)} + \eta d^{(k)}) &= \frac{1}{2} x^{(k)\top} Q x^{(k)} - b^\top x^{(k)} + \eta d^{(k)\top} Q x^{(k)} + \frac{1}{2} \eta^2 d^{(k)\top} Q d^{(k)} - \eta b^\top d^{(k)}, \quad b^\top d^{(k)} = d^{(k)\top} b \\ &= f(x^{(k)}) + \eta d^{(k)\top} (Q x^{(k)} - b) + \frac{1}{2} \eta^2 d^{(k)\top} Q d^{(k)} \end{aligned}$$

$$\begin{aligned} \nabla_\eta f(x^{(k)} + \eta d^{(k)}) &= d^{(k)\top} (Q x^{(k)} - b) + \eta d^{(k)\top} Q d^{(k)} = d^{(k)\top} \nabla_x f(x^{(k)}) + \eta d^{(k)\top} Q d^{(k)} \\ &= -d^{(k)\top} d^{(k)} + \eta d^{(k)\top} Q d^{(k)} = -\|d^{(k)}\|_2^2 + \eta d^{(k)\top} Q d^{(k)} \end{aligned}$$

$$\nabla_\eta f(x^{(k)} + \eta d^{(k)}) = 0 = -\|d^{(k)}\|_2^2 + \eta d^{(k)\top} Q d^{(k)} \quad \therefore \quad \eta = \frac{\|d^{(k)}\|_2^2}{(d^{(k)\top} Q d^{(k)})^{-1}}$$

1.2 LQR as a QP. Consider the Linear Time-Invariant (LTI) dynamical system

$$x_{t+1} = Ax_t + Bu_t,$$

where $A \in \mathbb{R}^{n \times n}$ and $B \in \mathbb{R}^{n \times m}$ are given matrices, and $x_t \in \mathbb{R}^n$ and $u_t \in \mathbb{R}^m$ are the system state and applied control input, respectively, at time $t \in \mathbb{N}_{\geq 0}$.

Let $x_0 \in \mathbb{R}^n$ be the fixed initial state and $T \in \mathbb{N}$ be some time horizon. Our goal is to find a sequence of control inputs $u^* := (u_0^*, u_1^*, \dots, u_{T-1}^*) \in \mathbb{R}^{mT}$ that minimizes the quadratic cost

$$J(u) := x_T^\top Q_T x_T + \sum_{t=0}^{T-1} (x_t^\top Q x_t + u_t^\top R u_t),$$

where $Q_T \in \mathbb{R}^{n \times n}$, $Q \in \mathbb{R}^{n \times n}$, and $R \in \mathbb{R}^{m \times m}$ are positive-definite matrices. Later, we will see how dynamic programming can be used to derive an elegant, recursive solution to this problem. For now, we study a convex least-squares formulation. Specifically, we reformulate the problem of minimizing $J(u)$ as

$$\min_{u \in \mathbb{R}^{mT}} \frac{1}{2} u^\top \tilde{Q} u - \tilde{b}^\top u,$$

where $u := (u_0, u_1, \dots, u_{T-1}) \in \mathbb{R}^{mT}$ is the vector of stacked control inputs, $\tilde{Q} \in \mathbb{R}^{mT \times mT}$ is a positive-definite matrix, and $\tilde{b} \in \mathbb{R}^{mT}$.

- (a) Write down $\tilde{Q}$ and $\tilde{b}$ in terms of Q_T , Q , R , A , B , and x_0 .
- (b) With this reformulation, implement the gradient descent algorithm of your choice to compute the optimal sequence of control inputs u^* for

$$Q_T = 10I_2, \quad Q = I_2, \quad R = I_1, \quad A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \quad x_0 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad T = 20,$$

where I_n is the identity matrix with dimension n . What is the optimal cost $J(u^*)$?

(a) Write down $\tilde{Q}$ and $\tilde{b}$ in terms of Q_T, Q, R, A, B , and x_0 .

$$X_1 = AX_0 + BU_0$$

$$X_2 = A(AX_0 + BU_0) + BU_1 = A^2X_0 + ABU_0 + BU_1$$

$$X_3 = A^3X_0 + A^2BU_0 + ABU_1 + BU_2$$

$$X_4 = A^4X_0 + A^3BU_0 + A^2BU_1 + ABU_2 + BU_3$$

↳

$$X = Vx_0 + WU$$

$$V = [A \ A^2 \ A^3 \ \dots \ A^T]^T, \quad X = [X_1^T, \dots, X_T^T]^T, \quad U = [U_0^T, \dots, U_{T-1}^T]^T$$

$$V = \begin{bmatrix} B & 0 & 0 & \dots & 0 \\ AB & B & 0 & \dots & 0 \\ A^2B & AB & B & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ A^{T-1}B & A^{T-2}B & A^{T-3}B & \dots & B \end{bmatrix}$$

$$\begin{aligned} J &= x_0^T Q x_0 + x_1^T Q_1 x_1 && + U_0^T R U_0 && T=1 \\ &= x_0^T Q x_0 + x_1^T Q x_1 + x_2^T Q_2 x_2 && + U_0^T R U_0 + U_1^T R U_1 && T=2 \\ &= x_0^T Q x_0 + x_1^T Q x_1 + x_2^T Q_2 x_2 + x_3^T Q_3 x_3 && + U_0^T R U_0 + U_1^T R U_1 + U_2^T R U_2 && T=3 \end{aligned}$$

$\underbrace{\hspace{15em}}_{X^T H X} \qquad \underbrace{\hspace{15em}}_{U^T N U}$

$$H = \text{diag}(Q, Q, \dots, Q_T), \quad N = \text{diag}(R, \dots, R)$$

$$J(U) = X^T H X + U^T N U$$

$$= (Vx_0 + WU)^T H (Vx_0 + WU) + U^T N U$$

$$= (Vx_0)^T H (Vx_0) + 2(Vx_0)^T H (WU) + (WU)^T H (WU) + U^T N U$$

$$= x_0^T V^T H V x_0 + 2x_0^T V^T H W U + U^T W^T H W U + U^T N U$$

$$= U^T (W^T H W + N) U + 2x_0^T V^T H W U + \underbrace{x_0^T V^T H V x_0}_{\text{not dependent on } U, \text{ ignored}}$$

$$= U^T (W^T H W + N) U + 2U^T V^T H W x_0 + \underbrace{x_0^T V^T H V x_0}_{\text{not dependent on } U, \text{ ignored}}$$

So, in the form $\frac{1}{2} U^T \tilde{Q} U - \tilde{b}^T U$:

$$\tilde{Q} = 2(W^T H W + N), \quad \tilde{b}^T = -2V^T H W x_0$$

1.3 Extremal curves. Given the functional

$$J(x) = \int_0^1 \left(\frac{1}{2} \dot{x}(t)^2 + 5x(t)\dot{x}(t) + x(t)^2 + 5x(t) \right) dt,$$

find an extremal curve $x^* : [0, 1] \rightarrow \mathbb{R}$ that satisfies $x^*(0) = 1$ and $x^*(1) = 3$.

Euler equation: $g_x(x^*, \dot{x}^*, t) - \frac{d}{dt} g_{\dot{x}}(x^*, \dot{x}^*, t) = 0$

$$g(x, \dot{x}, t) = \frac{1}{2} \dot{x}^2 + 5x\dot{x} + x^2 + 5x$$

$$\left. \begin{aligned} \frac{\partial}{\partial x} g(x, \dot{x}, t) &= 5\dot{x} + 2x + 5 \\ \frac{\partial}{\partial \dot{x}} g(x, \dot{x}, t) &= \dot{x} + 5x \end{aligned} \right\} \begin{aligned} 5\dot{x} + 2x + 5 - \frac{d}{dt}(\dot{x} + 5x) &= 0 \\ \cancel{5\dot{x}} + 2x + 5 - \dot{x} - \cancel{5\dot{x}} &= 0 \quad \therefore -\ddot{x} + 2x + 5 = 0 \\ \ddot{x} - 2x &= 5 \end{aligned}$$

Non-Homogeneous diff. equation

$$x(t) = x_h(t) + x_p(t)$$

$$r^2 - 2 = 0 \quad \therefore r = \pm \sqrt{2}$$

$$x_h(t) = C_1 e^{\sqrt{2}t} + C_2 e^{-\sqrt{2}t}$$

$$x_p(t) = C = -\frac{5}{2}$$

$$\dot{x}_p = 0, \quad \ddot{x}_p = 0 \quad \rightarrow 0 - 2C = 5 \quad \therefore C = -\frac{5}{2}$$

$$x(t) = C_1 e^{\sqrt{2}t} + C_2 e^{-\sqrt{2}t} - \frac{5}{2}$$

Using boundary conditions:

$$x(0) = 1 = C_1 e^{\sqrt{2} \cdot 0} + C_2 e^{-\sqrt{2} \cdot 0} - \frac{5}{2} = C_1 + C_2 - \frac{5}{2} \quad \therefore C_1 = \frac{7}{2} - C_2$$

$$x(1) = 3 = \left(\frac{7}{2} - C_2 \right) e^{\sqrt{2}} + C_2 e^{-\sqrt{2}} - \frac{5}{2} = \frac{7}{2} e^{\sqrt{2}} + (e^{-\sqrt{2}} - e^{\sqrt{2}}) C_2$$

$$C_2 = \frac{11 - 7 e^{\sqrt{2}}}{2(e^{-\sqrt{2}} - e^{\sqrt{2}})}, \quad C_1 = \frac{7}{2} - \frac{11 - 7 e^{\sqrt{2}}}{2(e^{-\sqrt{2}} - e^{\sqrt{2}})}$$

1.4 Zermelo's ship. Zermelo's ship must travel through a region of strong currents. The position of the ship is denoted by $(x(t), y(t)) \in \mathbb{R}^2$. The ship travels at a constant speed $v > 0$, yet its heading $\theta(t)$ can be controlled. The current moves in the positive x -direction with speed $w(y(t))$. The equations of motion for the ship are

$$\begin{aligned}\dot{x}(t) &= v \cos \theta(t) + w(y(t)) \\ \dot{y}(t) &= v \sin \theta(t)\end{aligned} \quad \dot{\mathbf{x}} = \mathbf{f}(\dots)$$

We want to control the heading $\theta(t)$ such that the ship travels from a given initial position $(x(t_0), y(t_0)) = (x_0, y_0)$ to the origin $(0, 0)$ in minimum time.

- (a) Suppose $w(y(t)) = \frac{v}{h}y(t)$, where $h > 0$ is a known constant. Show that an optimal control law $\theta^*(t)$ must satisfy a linear tangent law of the form

$$\tan \theta^*(t) = \alpha - \frac{v}{h}t$$

for some constant $\alpha \in \mathbb{R}$.

- (b) Suppose $w(y(t)) \equiv \beta$ for some constant $\beta > 0$. Derive an expression for the optimal transfer time $t_1^* - t_0$. Make sure to consider each of the following cases: (i) $v > \beta$, (ii) $v = \beta$, and (iii) $v < \beta$. State restrictions, if any, on x_0 under these conditions.

$$J = \int_{t_0}^{t_f} 1 \, dt$$

$$\dot{\mathbf{x}} = v \cos \theta + \frac{v}{h} y$$

a) Necessary Condition for Opt:

$$H = 1 + p_x \left[v \cos \theta + \frac{v}{h} y \right] + p_y v \sin \theta$$

$$\dot{x}^* = \frac{\partial H}{\partial p_x} = v \cos \theta + \frac{v}{h} y$$

$$\dot{y}^* = \frac{\partial H}{\partial p_y} = v \sin \theta$$

$$\dot{p}_x = -\frac{\partial H}{\partial x} = 0 \therefore p_x = c_1$$

$$\dot{p}_y = -\frac{\partial H}{\partial y} = -p_x \frac{v}{h} = -c_1 \frac{v}{h} \therefore p_y = -c_1 \frac{v}{h} t + c_2$$

$$0 = \frac{\partial H}{\partial \theta} = -p_x v \sin \theta + p_y v \cos \theta \therefore p_x \sin \theta = p_y \cos \theta \therefore \tan \theta^* = \frac{p_y}{p_x} = -\frac{v}{h} t + \frac{c_2}{c_1}$$

$$H = 1 + p_x \left[v \cos \theta + \beta \right] + p_y v \sin \theta$$

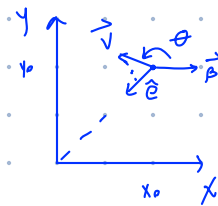
$$\dot{p}_x = -\frac{\partial H}{\partial x} = 0 \therefore p_x = c_1$$

$$\dot{p}_y = -\frac{\partial H}{\partial y} = 0 \therefore p_y = c_2$$

$$0 = \frac{\partial H}{\partial \theta} = -p_x v \sin \theta + p_y v \cos \theta \therefore p_x \sin \theta = p_y \cos \theta \therefore \tan \theta^* = \frac{p_y}{p_x} = \frac{c_2}{c_1}$$

$$\dot{x} = v \cos \theta + \beta$$

$$\dot{y} = v \sin \theta$$



b) Opt. time $\rightarrow$ max. speed to reach origin

$$t_f^* - t_0 = \frac{r_0}{s}, \quad r_0 = \sqrt{x_0^2 + y_0^2}$$

$$\hat{e} = \left(-\frac{x_0}{r_0}, -\frac{y_0}{r_0} \right)$$

For opt. time, need largest speed in $\hat{e}$ direction.

$$s = \vec{v} \cdot \hat{e} + \vec{\beta} \cdot \hat{e} = \vec{v} \cdot \hat{e} - \beta \frac{x_0}{r_0}$$

max when $\vec{v}$ is in the same direction of $\hat{e}$

$$s = (v - \beta \frac{x_0}{r_0}) \hat{e}$$

$$t_f^* - t_0 = \frac{r_0}{v - \beta \frac{x_0}{r_0}} = \frac{r_0^2}{v r_0 - \beta x_0}$$

i) $v > \beta$

$$\text{Since } \frac{x_0}{r_0} \leq 1 \quad \therefore x_0 \leq r_0, \quad v - \beta > 0$$

$$-\beta x_0 \geq -\beta r_0$$

$$v r_0 - \beta x_0 \geq v r_0 - \beta r_0 = r_0 (v - \beta)$$

$$v r_0 - \beta x_0 \geq r_0 (v - \beta) > 0 \quad \therefore v r_0 - \beta x_0 > 0$$

So denominator is always larger than zero. No restriction on x_0

ii) $v = \beta$

$$v - \beta = 0, \quad x_0 \leq r_0$$

$$v r_0 - \beta x_0 \geq r_0 (v - \beta) = 0$$

Not reachable when denominator is zero. This happens iff $r_0 = x_0 \Rightarrow y_0 = 0$
 $x_0 \geq 0$

iii) $v < \beta$

$$v - \beta < 0, \quad x_0 \leq r_0$$

$$v r_0 - \beta x_0 \geq r_0 (v - \beta) < 0 \quad \therefore v r_0 - \beta x_0 > 0$$

Origin reachable if $x_0 < \frac{v}{\beta} r_0$